

浙江大学



Lab Practice Report for Undergraduate

Name:

College: ZheJiang University

Department: Science

Major: Marine Science

Student NO.:

Instructor: Yanan Di

Date: 2024.12.11

Lab Practice Report of Zhejiang University

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Course Name: Marine biology Type of Practice: Produced

Name of Practice: Quantitative confirmation of microalgae

Student Name: _____ Major: Marine biology

_____ Student Number:

Team Memeber: Zhiang Liu

Instructor: Yanan Di

Practice Location: 241 Practice Date: 2024.10.10

1. Aims and Requirement of the Practice (Compulsory)

1. Learn the methods to reserve biological sample for long-term.
2. Practice the biological specimen production procedure.

2. Background information of the Practice

Plastination is the method of long term preservation of the biological tissues with completely visible surface and high durability. In this process, the water and lipids in biological tissues are replaced by curable polymers and the anatomical specimens are preserved by impregnation with these polymers which are subsequently hardened. Most often used polymers are the silicone, epoxy or polyester resins which keep the anatomical specimens in a dry, odourless state and they require minimal aftercare.

Plastination with polyester method will be practical. The polyester resins are a two part system, consisting of the base resin and the hardener. When mixed, a chemical reaction occurs where they go cure and harden. The drying time can highly vary from 12 to 48 hours.

3. Equipment and materials

1. Polyester casting resin: Resin A and Resin B
2. Prepared biological samples (dried)
3. Silicone mould
4. Silicone mixing cup, stirring stick, balance, Markers
5. Marker pen, gloves, paper towels

4.

5. Procedures

1. Pick up the biological samples and silicon mould you preferred

2. Calculate the total volume of resin required depending on the mould you selected.

3. Prepare the resin as the half of volume you have calculated: Mix the resin A and B in mixing cup following the volume ration of 2.5:1. Stir the mixture well with stirring stick by one direction until it becomes clear and clean (10 min will be enough). Please avoid the bubbles

4. Pour the mixture of resin into your mould. Make sure the filled volume is about 1/3 of the total volume of your mould.

5. Place your prepared sample in a clean place and waiting for it to be solid to provide the support for your biological samples (24 h used, as the samples is heavier than the float power the liquid resin can provide, therefore, solid form is required).

6. (Next day) Prepare the resin as procedure 4, the volume should be enough to top-up the mould.

7. Place your selected sample on the solid resin, carefully pour the freshly prepared resin into the mould and cover the sample without generating any bubble.

8. Place your sample in a clean place and waiting for another 24-48 h for complete cure and harden.

9. When the curing complete, you can now remove the mould and get long-term specimen for marine organism.

10. The specimen will be the souvenir of the course of "Marine Biology and Practice".

6. Questions

(1) Is there any other method to preserve organism for long-term, take at list one example.

Liquid nitrogen is quick-frozen and stored in a -80°C freezer.

Soak in formalin solution for preservation

(2) Take a picture of your preserved specimen and add in the report as an attachment.



(3)Score your prepared specimen as well as the specimen prepared by your team members (1-10 form poor to excellent), explain the reason to obtain such score.

10. I think my work is like a work of art. It's so perfect that it's definitely worth a 10 score.

10 Because his works have a sense of simplicity and simplicity.It is also a work of art

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